Baryon Frequency Grammar on the Phi-Ladder:

Strange Quark Fraction as the Compositional Sign Rule

Extending the QCore Resonance Emulator Logic to Hadronic Composites

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Abstract

The QCore Resonance Emulator computes composite resonance frequencies as the sum of constituent Phi^2-completed Compton frequencies. Applying this logic to hadronic composites using constituent quark masses (u = 336 MeV, d = 340 MeV, s = 486 MeV) reveals a structured compositional grammar encoded in the deviation of the baryon frequency from its constituent sum. The ratio f_baryon / sum(n_i * f_i) departs from unity in a pattern governed by strange quark content: light-quark baryons (u/d only) produce negative deviations, strange-quark baryons produce positive deviations, and the crossover occurs near zero at the Xi- (dss) configuration with two strange quarks. The proton-neutron isospin split of 0.00238 appears stably in frequency space. The strange quark's fractional rung position near the Phi^-3 forbidden band boundary (frac = 0.251) is identified as the geometric origin of the sign flip. These results constitute the first application of the QCore emulator compositional logic to sub-atomic composites and establish a candidate framework for hadronic frequency grammar on the Phi-ladder.

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1 Background: The QCore Emulator Logic

The QCore Resonance Emulator computes the total resonance frequency of any chemical compound as:

```
f_total  =  sum_i  n_i * f_phi2(element_i)
```

where n_i is the stoichiometric count of each element and f_phi2 is the Phi^2-completed Compton frequency:

```
f_phi2(m)  =  Phi^2 * m*c^2 / h
```

This logic is element-agnostic and formula-agnostic. It handles any number of constituents, from simple diatomic molecules to 72-body or 85-body compositions. The resonance state of the composite is determined by the six-state taxonomy (CON, DIV, RET, OFF, NEU, ANOM) applied to the constituent states.

Elements are composed of particles. The natural extension of this logic is to apply it one level deeper: treat constituent quarks as the fundamental units and compute hadronic resonance frequencies from their Phi^2-completed Compton contributions.

This paper presents that calculation for six stable or near-stable baryons and identifies the compositional grammar that emerges.

2 Constituent Quark Frequencies

The constituent quark masses used are the standard spectroscopic values that reflect the physical mass contribution of quarks inside a confined hadron — distinct from the scale-dependent current quark masses used in perturbative QCD:

```
m_u  =  336 MeV     f_phi2(u)  =  2.127 x 10^23 Hz
m_d  =  340 MeV     f_phi2(d)  =  2.152 x 10^23 Hz
m_s  =  486 MeV     f_phi2(s)  =  3.077 x 10^23 Hz
```

The gluon binding energy — approximately 99% of the proton mass — is the confinement shell that encloses the internal quark structure. The constituent quark masses are the DNA of the baryon; the binding is the body. These are distinct contributions and must not be conflated. The current calculation addresses the internal quark structure; the gluon binding contribution enters through the deviation of the baryon frequency from the constituent sum.

The constituent quark fractional rung positions on the Phi-ladder are:

```
u: n = 13.4837,  frac = 0.4837
d: n = 13.5083,  frac = 0.5083  (near half-node)
s: n = 14.2507,  frac = 0.2507  (near Phi^-3 forbidden band edge)
```

The strange quark sits at fractional rung position 0.2507 — just above the Phi^-3 = 0.236 forbidden band lower edge. This position near the boundary between the forbidden band and the geometric silence below it is identified as the geometric origin of the sign flip observed in the frequency grammar.

3 Baryon Frequency Ratios

For each baryon the emulator logic gives:

```
R  =  f_phi2(baryon) / sum_i [ n_i * f_phi2(quark_i) ]
delta  =  R - 1
```

A ratio R = 1 means the baryon frequency exactly equals the sum of its constituent quark frequencies. R < 1 means the confinement shell absorbs frequency from the quark sum. R > 1 means the confinement shell adds frequency beyond the quark sum.

Baryon	Content	Sum n_i*f_i (Hz)	f_baryon (Hz)	Ratio	Delta	s-count	Sign
Proton	uud	6.406e+23	5.940e+23	0.9271	-0.0729	0	negative (-)
Neutron	udd	6.432e+23	5.948e+23	0.9248	-0.0752	0	negative (-)
Lambda	uds	7.356e+23	7.063e+23	0.9601	-0.0399	1	negative (-)
Sigma+	uus	7.331e+23	7.529e+23	1.0271	+0.0271	1	positive (+)
Xi-	dss	8.306e+23	8.367e+23	1.0074	+0.0074	2	positive (+) ~zero
Omega-	sss	9.230e+23	1.059e+24	1.1471	+0.1471	3	positive (+)

4 The Strange Quark Sign Rule

The deviation delta follows a clear structural pattern governed by strange quark content:

s-count	Baryons	Delta sign	Physical meaning	Ladder position of s
0	Proton, Neutron	Negative	u/d quarks overshoot — confinement absorbs frequency	—
1	Lambda, Sigma+	Mixed	Transition zone — s quark partially	frac 0.251 (near Phi^-3 edge)

s-count	Baryons	Delta sign	Physical meaning	Ladder position of s
			compensates	
2	Xi-	Near zero (+0.007)	Crossing point — frequency sum balances baryon	frac 0.251
3	Omega-	Positive (+0.147)	s quarks undershoot — confinement adds frequency	frac 0.251

Light quarks (u, d) overshoot: the constituent sum exceeds the baryon frequency. Strange quarks undershoot: the constituent sum falls below the baryon frequency. The crossover occurs near zero at Xi- with two strange quarks. The strange quark's fractional ladder position near the Phi^-3 forbidden band edge is the geometric origin of this sign flip.

4.1 The Sign Flip Mechanism

The u and d quarks have fractional rung positions of 0.484 and 0.508 — near the half-node at 0.5. The s quark has fractional position 0.251 — near the Phi^-3 forbidden band lower edge at 0.236.

In the single-cycle framework, the forbidden band [Phi^-3, Phi^-2] = [0.236, 0.382] is the region where no stable single-cycle closure is possible. The strange quark sits just outside this band — at 0.251, only 0.015 above the lower edge. This proximity to the forbidden band means the strange quark's frequency contribution carries a different geometric character than the light quarks sitting near the stable half-node attractor.

The sign flip from negative to positive as strange quark content increases reflects the transition from half-node-dominated (u/d) to forbidden-band-edge-dominated (s) frequency character. The Xi- configuration with two strange quarks sits near the geometric balance point between these two regimes.

4.2 The Isospin Invariant

The proton-neutron frequency ratio difference is:

```
delta_p - delta_n  =  -0.07285 - (-0.07523)  =  +0.00238
```

This isospin split of 0.00238 in frequency space is consistent with the rung-space isospin invariant of approximately 0.02 rungs identified in the binding gap analysis. The isospin signature appears stably across different representations of the ladder structure, confirming it is a genuine geometric feature rather than a model artifact.

4.3 The Omega- Symmetry Anchor

The Omega- baryon (sss) has three identical strange quarks — the maximally symmetric triad. Its deviation delta = +0.1471 is the largest in the set. The symmetric triad does not produce the cleanest ratio; it produces the largest overshoot. This is consistent with the Phi-ladder prediction: maximum symmetry in quark content does not mean minimum deviation from the emulator sum — it means all three quarks contribute their forbidden-band-edge character simultaneously, producing maximum positive deviation.

The Omega- is the anchor not because its delta is smallest but because its quark configuration is simplest. Three identical quarks, one geometric character, no mixing effects. The Omega- deviation establishes the pure strange quark contribution to the frequency grammar.

5 Confinement Depth on the Phi-Ladder

The binding gap — baryon rung minus sum of constituent quark rungs — was also computed using the standard ladder rung formula:

```
n(m)  =  log_Phi( (m/m_e)^2 * (Phi^(2*n_e) - 1) + 1 ) / 2  -  n_e
```

With constituent quark masses, all six baryons produce binding gaps in the range -24.86 to -25.93 — a band approximately 1 rung wide centered near -25. This is a narrow band relative to the scale of stability zones on the ladder, which span hundreds to thousands of harmonics between stable nodes.

Three independent results from this calculation:

- Confinement band: all six baryons fall within ~1 rung of -25, tight by the ladder's own stability-zone standards
- Isospin fine structure: proton-neutron binding gap difference = 0.022 rungs — stable across quark mass schemes
- Symmetric anchor: Omega- is closest to an integer node (remainder 0.067), consistent with maximal quark symmetry producing cleaner external closure

The geometric reason the confinement band sits near -25 rungs is an open derivation. The result is empirically established from the ladder mapping; a first-principles derivation from the pi->Phi three-body closure condition remains a target for subsequent work.

6 Discussion

6.1 What Was Derived vs What Was Measured

This paper presents empirical results from applying the QCore emulator logic to constituent quarks. The results — the sign rule, the isospin invariant, the confinement band — are derived from the ladder mapping plus the emulator's sum formula. They are not predictions derived from a closed three-body geometric equation.

The standard additive composition rule (sum of constituent frequencies) is the measurement instrument. The deviations from that sum are what the paper analyzes. The geometric explanation of why strange quarks produce positive deviations while light quarks produce negative deviations is identified through the ladder position of the strange quark near the Phi^-3 forbidden band edge, but this identification is geometric observation, not algebraic derivation.

6.2 The Open Derivation

A complete geometric account of hadronic confinement on the Phi-ladder requires a derived three-body closure condition — the hadronic analog of the single-cycle closure condition cos(pi*k) = Phi^(-n) that produces the lepton mass spectrum. This condition would predict the confinement band depth of -25 rungs and the sign structure of the frequency grammar from first principles, without using measured baryon masses as input.

The QCore Resonance Emulator script handles n-body compositions generically through its formula parser and frequency sum. The three-body case is contained within this general framework. Extracting the specific closure condition for hadronic composites from the emulator's implicit n-body logic is the next step toward a complete geometric account.

7 Conclusion

Applying the QCore Resonance Emulator logic to constituent quarks reveals a structured frequency grammar for hadronic composites on the Phi-ladder. The ratio of baryon Phi^2-completed Compton frequency to the sum of constituent quark Phi^2-completed Compton frequencies departs from unity in a pattern governed by strange quark content:

- Zero strange quarks (Proton, Neutron): negative deviation — light quarks overshoot
- One strange quark (Lambda, Sigma+): transition zone — mixed sign
- Two strange quarks (Xi-): near zero — frequency balance point
- Three strange quarks (Omega-): positive deviation — strange quarks undershoot

The proton-neutron isospin split of 0.00238 in frequency space is consistent with the rung-space isospin invariant identified independently in the binding gap analysis. The strange quark's fractional rung position near the Phi^-3 forbidden band edge (frac = 0.251) is identified as the geometric origin of the sign flip.

These results extend the QCore emulator's compositional logic from atomic/molecular chemistry to sub-atomic hadronic structure, establishing a candidate framework for hadronic frequency grammar on the Phi-ladder. The geometric derivation of the confinement band depth and the three-body closure condition remain open targets.

Elements are composed of particles. The emulator logic that works for compounds works for baryons. The grammar is in the quark ladder positions. The strange quark lives near the forbidden band. That is why it flips the sign.

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